

# Ankit Pokhrel

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## Summary

**Computer Engineering** undergraduate specialising in **Artificial Intelligence** and **Machine Learning**, with hands-on experience building **end-to-end ML and deep learning systems**, from data preprocessing, feature engineering, and model training to **NLP, RAG, and LLM applications** deployed in production.

## Technical Skills

**Languages** : Python, C, C++, JavaScript, TypeScript, SQL

**AI/ML Concepts** : Deep Learning, CNNs, RNN/LSTM/GRU, Transformers & Attention, NLP, RAG, LLM Fine-Tuning (QLoRA/PEFT), Computer Vision, Model Evaluation, Feature Engineering

**ML & Data** : PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face, LangChain, NumPy, Pandas, OpenCV, librosa, Matplotlib

**Backend & Web** : FastAPI, Flask, Node.js, Express, React.js, React Native, REST APIs

**Databases** : PostgreSQL, MongoDB, Firebase, ChromaDB, pgvector, Redis

**Tools & MLOps** : Docker, Git, GitHub, Coolify, Streamlit, Jupyter, Prisma

## Experience

### AI/ML Intern

May 2026 – Present

*Layman AI, Remote*

- Built a **sports analytics pipeline** using **YOLOv8** for player/object detection, **MediaPipe** for pose estimation, and rule-based shot classification, producing annotated video output with per-frame **CSV/JSON** analytics.
- Developed and exposed **RESTful APIs** with **FastAPI** to serve real-time computer vision inference, integrating detection pipelines with live application streams.
- Contributed to production-grade CV systems aligned with the company's real-time analytics products (parking occupancy, construction safety, activity monitoring), delivering features on schedule within an agile team.

### Secretary, IT Club Pulchowk

June 2025 – June 2026

*Lalitpur, Nepal*

- Deployed and maintained campus student and college web projects on a self-hosted **Manaslu server** using **Coolify** (PaaS), managing **Docker**-based deployments, subdomains, and service uptime.
- Organized technical workshops, seminars, and hackathons for engineering students.
- Oversaw development and maintenance of the IT Club website and digital assets.

### Fellow, Seeds for Future

June 2025 – July 2025

*Nepal*

- Completed intensive training on **AI, 5G, Cloud Computing**, and digital infrastructure.
- Worked on labs involving AI applications and cloud-based systems.

## Education

### Bachelors in Computer Engineering

April 2023 – Present

*IOE, Pulchowk Campus*

*Lalitpur, Nepal*

## Projects

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### Network Anomaly Detection in Named Data Networking     *Python, scikit-learn, ndnSIM*     *Ongoing Research*

- Building an **unsupervised anomaly-detection system** (**Isolation Forest**, scikit-learn) under **Prof. Babu Ram Dhawadi** to detect **Interest Flooding** and **Cache Pollution** attacks in **Named Data Networking (NDN)** across 3 topologies from unlabeled normal traffic.
- Achieved **100% attacker detection** with **~1s latency** and **0% false-positive rate**; validated on two platforms (ns-3/**ndnSIM** simulation and live **miniNDN/NFD** emulation).
- Applied **KS hypothesis testing** and **feature-ablation** studies to isolate attack-carrying features; diagnosed a data-quality issue that raised detection from **8.5% to 61%** with no model change.

### English–Nepali Neural Machine Translation (Transformer from Scratch)     *Python, PyTorch, NLP*     [Code](#)

- Implemented an **encoder–decoder Transformer (Vaswani et al. 2017) from scratch** in PyTorch (**multi-head self/cross-attention**, sinusoidal positional encoding, label smoothing, and warmup LR schedule) with no high-level NMT libraries.
- Trained a **35M-parameter model** on **1M sentence pairs** filtered from Meta AI’s 19.6M-pair NLLB corpus; scored **13.8 BLEU / 41.2 ChrF** on the standard **FLORES-200** benchmark.
- Optimized the data pipeline with **SentencePiece BPE** tokenization and length-bucketed batching, eliminating ~80% padding overhead and cutting epoch time **~12x**.

### Kanun Sathi – AI Legal Assistant (RAG)     *FastAPI, RAG, pgvector, Gemini*     [Source Code](#)

- Built a **Retrieval-Augmented Generation (RAG)** platform (team of 4) answering legal questions in **Nepali** with verifiable **page-level citations** to source law (Constitution, criminal/family law).
- Engineered the pipeline: PDF chunking, **768-dim embeddings** (Ollama), **pgvector** similarity search, and **Gemini 2.5 Flash** generation with citation-enforcing prompts; async ingestion via **Redis/ARQ** workers.
- Developed a **React + TypeScript** frontend and containerized the full stack (FastAPI, PostgreSQL/pgvector, Redis, Ollama) with **Docker Compose**.

### ragchatbot: Open-Source Python RAG SDK     *Python, FastAPI, ChromaDB*     [PyPI](#)

- Designed and **published an open-source, pip-installable RAG SDK (v1.1.3 on PyPI)** that turns a document folder into a production-ready semantic-search chatbot API with zero ML setup.
- Built a **pluggable multi-LLM layer** (Gemini, OpenAI, local Ollama) behind one interface, **ChromaDB** vector store with MD5-based incremental indexing, token streaming, and a **25-test pytest suite**.

### Nagarik Suchana: Civic-Tech Platform     *React Native, Node.js, PostgreSQL*     [Source Code](#)

- Built a **full-stack bilingual (English/Nepali)** civic app (React Native, Node.js/Express, PostgreSQL) connecting citizens with local government for notices, grievances, and governance transparency.
- Implemented an **OCR pipeline (Tesseract.js) + Gemini AI** to extract **30+ structured fields** from government notice images/PDFs for intelligent citizen–notice matching; containerized with Docker.

### Infant Cry Detection     *TensorFlow, LSTM, Audio Processing*     [Source Code](#)

- Built an end-to-end **audio classification** system (TensorFlow LSTM/ANN) classifying infant cries (**hunger, pain, discomfort**) using **MFCC** and spectral features on 1,000+ recordings, with **SMOTE** for class imbalance.
- Deployed via a **Flask REST API** and a **cross-platform React Native/Expo** app (iOS/Android/Web) using **TensorFlow Lite** for on-device inference.

## Achievements

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- **1st Runner-Up – National Hackathon (2025)**: Secured 1st Runner-Up position in the hackathon held alongside the **4th National Telemedicine and 1st National Digital Health Conference**, contributing innovative solutions in digital health and telemedicine.
- **Best 1st-Year Electronics Project Award**: Received certificate and cash prize for developing a **Parking Management System using Arduino**

## Certifications

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- [Machine Learning – Coursera](#)
- [Meta Front-End Development – Coursera](#)
- [Coding and Programming with Python – Samsung Innovation College](#)